DATA CLEANING - HANDLING MISSING VALUES

Take a first look at the data and how many missing data points do we have?

```
import pandas as pd
import numpy as np
# read in all our data
dataset = pd.read_csv("C:\\NFL Play by Play 2009-2017 (v4).csv")
# I can see a handful of missing data already!
print(dataset.head() )
print(dataset.shape[0])
print(dataset.shape[1])

# get the number of missing data points per column
missing_values_count = dataset.isnull().sum()

# look at the # of missing points in the first ten columns
print(missing_values_count[0:10])
print(missing_values_count.shape)

# how many total missing values do we have?
total_cells = np.product(dataset.shape)
total_missing = missing_values_count.sum()
print(total_cells)
print(total_missing)
```

```
# percent of data that is missing
percent_missing = (total_missing/total_cells) * 100
print(percent_missing, '%')

print (missing_values_count[4])

percent_missing_col0 = (missing_values_count[0]/dataset.shape[0]) * 100
print (percent_missing_col0 , '%')

percent_missing_col4 = (missing_values_count[4]/dataset.shape[0]) * 100
print (percent_missing_col4 , '%')
```

Figure out why the data is missing

Is this value missing because it wasn't recorded or because it doesn't exist?

There are other fields, like "PenalizedTeam" that also have lot of missing fields. In this case, though, the field is missing because if there was no penalty then it doesn't make sense to say *which* team was penalized. For this column, it would make more sense to either leave it empty or to add a third value like "neither" and use that to replace the NA's.